

សាកលវិទ្យាល័យភូមិន្ទភ្នំពេញ

ROYAL UNIVERSITY OF PHNOM PENH

ប្រភាគបទរចាយការណ៍ស្រាវជ្រាវ

Multi-classification of Harmful Images based-on Deep Convolutional Neural Networks and Transfer Learning

A Research Report

In Partial Fulfilment of the Requirement for the Degree of

Master of Science in Information Technology Engineering

ញ៉ែម សុជាតា

NHEM SOCHEATA

December 24

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Examination committee: ……

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December 2024

Abstract in Khmer

នៅក្នុងយុគសម័យសម័យបច្ចេកវិទ្យា ការកើនឡើងនូវបណ្តាញកម្មវិធីសង្គម ជាច្រើនរៀងរាល់ថ្ងៃ ដែលធ្វើអោយការចែកចាយនៅរាល់រូបភាពផ្សេងៗពាស់ពេញ អ៊ីនធឺណិតលើពិភពលោក។ ជាមួយការប្រើប្រាស់កម្មវិធីកាន់តែងាយស្រួលជាងពី មុនៗសម្រាប់បុគ្គលរៀងៗខ្លួនបាន ធ្វើឲ្យការចែករំលែករូបភាពរបស់អ្នកប្រើប្រាស់ទៅ កាន់កម្មវិធីផ្សេងៗយ៉ាងងាយស្រួល។ ទោះបីជាយ៉ាងណាក៏ដោយ ភាពងាយស្រួលនេះ ក៏បានចូលរួមចំណែកដោយចេតនា ក្នុងការផ្សព្វផ្សាយមាតិកា ឬរូបភាពមិនសមរម្យ ដែលទាក់ទងទៅនឹង ការផ្សព្វផ្សាយគ្រឿងស្រវឹង រូបភាពសុិចស៊ី រូបភាពទាក់ទងនឹង ឈាម គ្រោះថ្នាក់ចរាចណ៍ផ្សេងៗ។ មាតិកា ឬរូបភាពទាំងនោះមានខ្លឹមសារដែលបង្ក ហានិភ័យយ៉ាងធ្ងន់ធ្ងរដល់អ្នកប្រើប្រាស់បណ្តាញកម្មវិធីផ្សេងនៅលើអ៊ីធើណែត ជា ពិសេសក្រុមដែលងាយរងគ្រោះដូចជា កុមារ និងក្មេងជំទង់ ដែលអាចខ្វះភាពចាស់ទុំ ខាងផ្លូវអារម្មណ៍ ក្នុងការយល់ដឹង។ ការទទួលបានបាននៅមាតិកា ឬរូបភាពទាំងនោះ ដែលអាចនាំឲ្យមានបញ្ហាផ្លូវចិត្ត អារម្មណ៍មិនល្អ ឬថែមទាំងរារាំងដល់ការអភិវឌ្ឍន៍ របស់ពួកគេទៀតផង។

ដើម្បីប្រឆាំងនឹងបញ្ហាដែលកំពុងកើនឡើងនេះ បណ្តាញកម្មវិធីមួយចំនួនបាន ធ្វើការអនុវត្តប្រព្ធន័ត្រួតពិនិត្យដោយស្វ័យប្រវត្តិ ដើម្បីទប់ស្កាត់រាល់មាតិកា ឬរូបភាព ដែលបង្កគ្រោះថ្នាក់មុនពេលដាក់អោយអ្នកប្រើប្រាស់របស់ពួកគេ។ ផ្ទុយទៅវិញ នៅ មានបណ្តាញកម្មវិធីមួយចំនួនបាន ពឹងផ្អែកលើការត្រួតពិនិត្យដោយដៃផ្ទាល់ផងដែរ។ ខណៈពេលវិធីសាស្ត្រទាំងពីរ នេះមានគោលបំណង បង្កើតឡើងដើម្បីសុវត្ថិភាពដល់ អ្នកប្រើប្រាស់កម្មវិធីទាំងអស់នោះ។ ការត្រួតពិនិត្យងាយៗ ឬដោយដៃ គឺចំណាយ ពេលវេលាច្រើន និងកម្លាំងពលកម្មច្រើន ហើយថែមទាំងពេលខ្លះ មិនបានត្រួតពិនិត្យ ជាប់លាប់ផងដែរ។ យ៉ាងវិញទៀត ក៏អាចប៉ះពាល់អារម្មណ៍ដល់អ្នកត្រួតពិនិត្យ ដែល ត្រូវមើលរូបភាពទាំងនេះជាប្រចាំ។ ទន្ទឹងនឹងនេះការត្រួតពិនិត្យដោយប្រព្ធន័ស្វ័យប្រវត្តិ ក៏អាចមានភាពលំបាកលើការងារដែលស្មុគស្មាញ ភាពត្រឹមត្រូវ ឬពេលវេលា ភ្លាមៗ ទៅលើមាតិកា ឬរូបភាពទាំងនោះផងដែរ។

ការស្រាវជ្រាវនេះ ដើម្បីស្វែងរកនូវដំណោះស្រាយបញ្ហាប្រឈមទាំងនេះដោយបង្កើតនៅប្រព័ន្ធត្រួតពិនិត្យដោយស្វ័យប្រវត្តិដោយចាត់ថ្នាក់រូបភាពដែលបង្កគ្រោះថ្នាក់ដូចជាគ្រឿងស្រវឹង ឈាម ស៊ីចស៊ី និងគ្រោះថ្នាក់ចរាចរណ៍ជាដើម។ ជាស្នូលនៃប្រព័ន្ធនេះបានបង្កើតឡើងដោយប្រើនូវបច្ចេកទេស Deep Learning ។ ជាពិសេសបច្ចេកទេស Convolutional Neutral Network ដែលមានប្រសិទ្ធភាពខ្លាំង ធ្វើការជាមួយរូបភាពដោយស្វ័យប្រវត្តិ ហើយរៀនពីមុខងារនៃ Raw Pixel ទិន្នន័យ។ តាមរយប្រើប្រាស់បច្ចេកទេស Convolutional Neutral Network ដែលការស្រាវជ្រាវនេះបង្កើនប្រសិទ្ធភាព ក្នុងការចាត់ថ្នាក់រូបភាពដែលអនុញ្ញាតអោយបណ្តាញកម្មវិធីកត់ចំណាំនូវរូបភាពបានច្បាស់លាស់ នឹងអាចពង្រីកសមត្ថភាពប្រព័ន្ធនៅពេញក្រោយទៀតផងដែរ។

ជាក់ស្តែងការប្រើប្រាស់បច្ចេកទេស Transfer Learning អាចឲ្យប្រព័ន្ធទទួលបានអត្ថប្រយោន៍ពីម៉ូឌែលដែលបានបណ្តុះបណ្តាលទុកមុនដូចជា InceptionV3 ដែលត្រូវបានបង្រៀនរួចជាស្រេចលើសំណុំទិន្នន័យជាច្រើន។ បច្ចេកទេសនេះអាចឲ្យម៉ូឌែលរបស់យើងរៀនបានរហ័សជាងមុន ក្នុងការចាត់ថ្នាក់រូបភាពដែលបង្កគ្រោះថ្នាក់ទិន្នន័យមានដែនកំណត់។ បច្ចេកទេស Transfer Learning ជួយកាត់បន្ថយនូវធនធានកុំព្យូទ័រ និងពេលវេលា ក្នុងការបណ្តុះបណ្តាល អោយក្លាយជាវិធីសាស្ត្រ ដែលមានប្រសិទ្ធភាពជាងមុន។ ជាពិសេស នៅក្នុងប្រទេសកម្ពុជា ដែលបរិបទវប្បធម៌ដើរតួនាទីយ៉ាងសំខាន់ក្នុងការត្រួតពិនិត្យមាតិកាផ្សេងលើបណ្តាញមកម្មវិធី។

ដើម្បីបង្កើនសមត្ថភាពរបស់ប្រព័ន្ធ និងការពារកុំឲ្យដំណើរការលើស ក្នុងការបណ្តុះបណ្តាលម៉ូឌែល។ បច្ចេកទេសបង្កើនទិន្នន័យ ត្រូវការអនុវត្តដូចជា បង្វិលដោយចៃដន្យ ត្រឡប់ ឬពង្រីករូបភាព។ ការបំប្លែងទាំងនេះជួយពង្រីក សំណុំទិន្នន័យដោយសិប្បនិម្មិតដែលអនុញ្ញាតឲ្យម៉ូឌែលរៀនពីបច្ចេកទេសបំប្លែងទិន្នន័យនេះកាន់តែមានភាពទូលំទូលាយ។ ជាពិសេស ធ្វើឲ្យម៉ូឌែលមានសមត្ថភាពខ្លាំងជាងមុន ដោយធានាថាអាចគ្រប់គ្រង់គ្រប់ទម្រង់រូបភាពផ្សេងៗដូច ជាលក្ខខណ្ឌពន្លឺ នឹងការបង្ហួមរូបភាព។

ជាលទ្ធផលផងដែរ ការស្រាវជ្រាវនេះបានបង្ហាញថា ការប្រើប្រាស់បច្ចេកទេស Convolutional Neutral Network ជាមួយនឹង Transfer Learning ផ្តល់នូវដំណោះស្រាយដ៏មានប្រសិទ្ធភាពមួយ សម្រាប់ចាត់ថ្នាក់រូបភាព ដែលអាចបង្កគ្រោះថ្នាក់បានទៅដល់អ្នកប្រើប្រាស់កម្មវិធី។ បច្ចេកទេសនេះបាន បង្ហាញពីភាពត្រឹមត្រូវ និងប្រសិទ្ធភាពខ្ពស់

ជំនួសដោយការត្រួតពិនិត្យអោយបច្ចេកទេសធម្មតា។ ជាពិសេស សម្រាប់បណ្តាញកម្មវិធីណាដែលមានទំហំបង្ហោះរូបភាពជាច្រើន។ លើសពីនេះ ផ្តោតទៅលើមាតិកាដែលមានរូបភាពអាចបង្កគ្រោះថ្នាក់ ដែលបានបែងចែកថ្នាក់ច្បាស់លាស់ ដែលប្រព្ធន័នេះអាចសម្របខ្លួនទៅ នឹងបរិបទមាតិកា ក្នុងប្រទេសកម្ពុជាដែលកំពុងពេញនិយមដូចជា គ្រឿងស្រវឹង ឬគ្រោះថ្នាក់ចរាចរណ៍ ដែលទាមទារឲ្យមានការយកចិត្តទុកដាក់ខ្លាំងបំផុតផងដែរ។

ពាក្យគន្លឹះ: Social media, Deep Learning, Convolutional Neural Networks (CNNs), Image Classification, Transfer Learning, Harmful Content, Automated Moderation, Content Filtering, Cultural Sensitivity.

Abstract in English

In today's digital age, the rapid growth of social media platforms has significantly increased the sharing of images across various online spaces. With the rise of user-friendly applications, it has become easier than ever for individuals to share a wide array of images, ranging from personal photos to newsworthy events. However, this convenience has inadvertently contributed to the widespread dissemination of harmful content, including explicit images, scenes of violence, traffic accidents, and depictions of substance abuse. Such harmful content poses a serious risk to viewers, particularly vulnerable groups like children and adolescents who may lack the emotional maturity to fully understand or process such images. Exposure to inappropriate material can lead to psychological distress, emotional turmoil, and may even hinder their development.

To combat this growing issue, some platforms have implemented automated monitoring systems designed to filter harmful content before it reaches users. In contrast, other platforms still rely on manual content moderation, where human moderators review images to ensure they meet the platform's content guidelines. While both approaches aim to create safer online spaces, they face significant challenges in handling the vast volume of images shared daily. Manual moderation, for instance, is time-consuming, labor-intensive, and often inconsistent. It can also lead to emotional burnout for human moderators who must view disturbing images on a regular basis. Meanwhile, automated systems may struggle to maintain high levels of accuracy and efficiency, particularly when dealing with diverse, complex images in real time.

This research seeks to address these challenges by developing an automated image classification system that can accurately and efficiently detect and classify harmful images into specific categories, such as alcohol, blood, sexual content, and traffic accidents. The core of this system is built on deep learning techniques, specifically Convolutional Neural Networks (CNNs), which have shown remarkable performance in image-based tasks due to their ability to automatically learn hierarchical features from raw pixel data. By leveraging the power of CNNs, this study aims to improve the detection accuracy of harmful images, allowing platforms to identify inappropriate content more reliably and at scale.

A significant aspect of our approach is the use of transfer learning, which allows the system to benefit from pre-trained models like InceptionV3, which have already been

trained on large, diverse datasets. This technique enables the model to adapt quickly to the task at hand—harmful image classification—even with limited labeled data. Transfer learning significantly reduces the computational resources and time needed for training, making it a more efficient approach, particularly when dealing with smaller, specialized datasets like the one used in this study. Additionally, the model's architecture was fine-tuned to ensure high performance on categories that are particularly relevant to online platforms, especially in Cambodia, where cultural context plays an important role in content moderation.

To improve the system's generalization ability and prevent overfitting, data augmentation techniques were applied, including random rotations, flips, and zooms. These transformations artificially expand the dataset, allowing the model to learn from a wider range of image variations. This helps the model become more robust, ensuring that it can handle images with different orientations, lighting conditions, and distortions.

The results of this research suggest that CNNs, enhanced with transfer learning, offer an effective solution for real-time harmful content detection and classification. The system demonstrated high accuracy and efficiency, making it a viable alternative to manual content moderation, especially for platforms with large-scale image uploads. This approach also presents a scalable solution to meet the increasing demand for automated moderation in the digital age. Furthermore, by focusing on specific harmful content categories, this system can be adapted to meet the cultural and contextual needs of different regions, as evidenced in the Cambodian context, where certain types of harmful content—such as alcohol or traffic accidents—require focused attention.

Keywords: Social media, Deep Learning, Convolutional Neural Networks (CNNs), Image Classification, Transfer Learning, Harmful Content, Automated Moderation, Content Filtering, Cultural Sensitivity.

SUPERVISOR'S RESEARCH SUPERVISION STATEMENT

TO WHOM IT MAY CONCERN

Name of program: Master of Science in Information Technology Engineering
Name of candidate: Nhem Socheata

Title of research report: Multi-classification of Harmful Images based-on Deep Convolutional Neural Networks and Transfer Learning

This is to certify that the research carried out for the above titled master's research report was completed by the above-named candidate under my direct supervision. This thesis material has not been used for any other degree. I played the following part in the preparation of this research report:

Supervisor's name: Dr. Chhoeum Vantha

Supervisor's signature……

Date……

Acknowledgments

I would like to express my deepest gratitude to everyone who has supported and contributed to the completion of this research.

First and foremost, I am profoundly thankful to my advisor, **Dr. Chhoeum Vantha**, for his invaluable guidance, insightful feedback, and unwavering support throughout this journey. His expertise and encouragement have been instrumental in shaping this work and fostering my academic growth.

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To my friends, colleagues, and fellow students, I am deeply grateful for your camaraderie, support, and inspiration. Your encouragement and understanding made the challenges of graduate studies more manageable and enjoyable.

Finally, I would like to express my profound appreciation to my family for their love, patience, and unwavering belief in me. Their constant support has been the driving force behind both my academic and personal achievements.

Thank you all for your contributions to this journey.

Sincerely,
Nhem Socheata

CANDIDATE'S STATEMENT

TO WHOM IT MAY CONCERN

I, **Nhem Socheata**, am passionate about advancing research in the field of Deep Learning at the **Royal University of Phnom Penh**. My Master's degree has equipped me with robust research capabilities and a deep theoretical understanding, allowing me to explore and contribute to innovative projects in this domain. Through my work on projects such as Deep Learning, I have refined my analytical skills, honed my critical thinking abilities, and strengthened my commitment to conducting research with integrity and precision.

My long-term research focus is on addressing challenges in **Image Classification**, and I am confident that this research report marks a significant milestone toward achieving that goal. I am eager to continue contributing to the ongoing research efforts at the **Royal University of Phnom Penh** while further developing my expertise in this field.

I affirm that this research report, submitted for the degree of **Master of Science** at the **Royal University of Phnom Penh**, is entirely my own work. Furthermore, it has not been used to fulfill the requirements of any other qualification, in whole or in part, at this or any other university or equivalent institution.

No reference to, or quotation from, this document may be made without the written approval of the author.

Signed by (the candidate): ……

Date: …….

Sign by Supervisor: ……...

Supervisor's signature: ……..…

Date……...……

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CHAPTER 1
INTRODUCTION

1.1 Background to the Study

Image classification is a fundamental task in the field of computer vision, focusing on assigning specific labels or categories to images based on their visual content. Historically, traditional classification methods relied heavily on handcrafted feature extraction techniques, such as the Scale-Invariant Feature Transform (SIFT), Histogram of Oriented Gradients (HOG), and color histograms. While these approaches had certain merits, they often required extensive domain expertise and suffered from limited scalability and accuracy, particularly when dealing with complex or diverse datasets.

The introduction and advancement of deep learning, particularly Convolutional Neural Networks (CNNs), has significantly revolutionized the field of image classification. CNNs are specifically designed to automatically extract hierarchical visual features, capturing intricate patterns from raw pixel data without the need for manual feature engineering. This breakthrough has resulted in a substantial improvement in classification accuracy, making CNNs indispensable in a broad range of applications, including medical imaging, facial recognition, autonomous vehicles, and notably, digital content moderation systems.

In recent years, the explosive growth of digital platforms and social media has dramatically increased the volume and accessibility of user-generated content, especially visual content. While this has positively influenced social connectivity and information exchange, it has simultaneously created significant moderation challenges, particularly related to the proliferation of harmful or inappropriate images. Such harmful images can include violent scenes, explicit or sexually suggestive content, graphic injuries, substance abuse, and images depicting traumatic incidents like traffic accidents. The dissemination of such content can profoundly impact viewers, especially vulnerable populations such as children and adolescents, potentially leading to psychological trauma, desensitization, anxiety, and societal unrest.

Given the high stakes and large-scale nature of this issue, relying solely on traditional manual moderation methods—human reviewers screening images individually—has proven insufficient. Manual moderation faces several critical

limitations, including subjective bias, inconsistency, psychological harm to moderators, and the sheer impracticality of reviewing vast numbers of images uploaded daily. Therefore, there is an urgent demand for effective automated moderation systems capable of accurately and swiftly classifying harmful images to protect users and maintain community standards.

Deep learning methods, specifically CNNs combined with transfer learning, offer a highly promising solution to this challenge. Transfer learning significantly enhances the efficiency of developing powerful CNN models by leveraging pre-existing knowledge gained from training on large-scale datasets, such as ImageNet. Pre-trained CNN models like VGG, ResNet, Inception, and EfficientNet have demonstrated remarkable capabilities in capturing generic and complex visual features, which can then be adapted to specialized image classification tasks, even when datasets are relatively small or specific to a particular cultural or regional context.

In the context of Cambodia, where internet penetration and digital content sharing have rapidly expanded, localized and culturally sensitive moderation of harmful images is critically important. Automated multi-classification of harmful images, specifically targeting categories such as Alcohol, Blood, Sexual Content, and Traffic Accidents, is crucial due to their relevance and potential harm in the Cambodian digital context. However, multi-class classification of harmful images introduces several distinct challenges, such as class imbalance, overlapping visual characteristics between different harmful categories, interpretability of automated decisions, scalability for real-time processing, and ethical considerations in accurately labeling sensitive content.

Therefore, this research focuses explicitly on addressing these challenges through the development of a CNN-based automated system utilizing transfer learning (InceptionV3) to classify harmful images into clearly defined categories. This study aims to provide a scalable, accurate, and culturally responsive solution to moderating harmful content, significantly reducing reliance on manual moderation processes and ultimately contributing to safer digital environments, particularly within localized Cambodian contexts.

1.2 Problem Statement

The rapid growth in digital platforms and widespread adoption of social media have dramatically increased the volume and diversity of user-generated content. While this has

significantly improved communication and connectivity, it has also introduced substantial challenges in maintaining community standards, especially regarding harmful or inappropriate imagery. Traditional content moderation methods, which predominantly rely on manual human review, are increasingly inadequate to cope effectively with the scale, complexity, and speed of today's digital landscape.

One critical challenge is the overwhelming volume of visual content uploaded daily on platforms such as Facebook, TikTok, Telegram, and online forums, particularly in rapidly digitalizing regions like Cambodia. This unprecedented surge in content requires moderation systems to scale efficiently while maintaining high accuracy. In Cambodia specifically, rising internet usage has resulted in an increased amount of culturally specific visual content, posing unique moderation challenges that generic, global moderation systems often fail to address adequately.

Moreover, reliance on manual moderation processes places significant psychological and physical strain on human moderators, who must continuously review graphic, violent, explicit, or distressing content. Human moderators frequently experience fatigue, psychological distress, and burnout, reducing their efficiency and accuracy. Subjective biases and cultural differences further complicate the consistency of moderation decisions, resulting in the occasional misclassification or oversight of harmful imagery. This issue becomes even more severe during peak activity periods or local cultural events, where moderation resources become insufficient, and harmful content remains publicly accessible for extended periods.

Considering these substantial limitations, it has become increasingly necessary to transition toward advanced automated systems powered by artificial intelligence (AI), particularly those leveraging Deep Convolutional Neural Networks (CNNs). AI-driven moderation systems offer robust solutions capable of processing vast amounts of visual data efficiently and accurately. CNN-based models, enhanced by transfer learning techniques, have demonstrated potential for achieving high accuracy in categorizing harmful images into multiple clearly defined classes, such as Alcohol, Blood, Sexual Content, and Traffic Accidents.

However, the development of effective multi-class classification models faces several specific challenges, including class imbalance—where harmful content may constitute a minority within large datasets—ambiguity in classification due to overlapping visual

features among harmful image categories, interpretability of classification results, and scalability to perform real-time moderation. Addressing these challenges requires the integration of culturally relevant datasets and advanced methods such as data augmentation, regularization techniques, and explainable AI (XAI).

Therefore, this study addresses these critical moderation issues by developing a culturally sensitive, accurate, and scalable automated harmful image classification system. Using a CNN-based model combined with transfer learning (specifically InceptionV3), this research aims to overcome current moderation limitations, minimize exposure to harmful content, and significantly improve the efficiency and reliability of digital content moderation, particularly within the Cambodian context.

1.3 Aim and Objectives of the Study

The primary aim of this research is to design and implement an deep learning model for classifying harmful images into four specific categories: Alcohol, Blood, Sexual Content, and Traffic Accidents. The proposed solution utilizes convolutional neural networks (CNN) and transfer learning techniques to achieve scalability, accuracy, and culturally relevant moderation of inappropriate visual content, with a particular focus on addressing local moderation challenges within the Cambodian digital environment.

The research specifically seeks to curate a diverse, culturally relevant, and representative dataset consisting of approximately 4,000 images, clearly categorized into four classes:

- Alcohol: Images illustrating alcohol consumption, bottles, intoxicated individuals, or alcohol-related paraphernalia.
- Blood: Images depicting violence, physical injuries, bloodstains, or medical emergencies involving visible blood.
- Sexual Content: Images containing explicit nudity, sexually suggestive material, or sexually explicit symbols.
- Traffic Accidents: Images depicting vehicle accidents, damaged vehicles, injured persons, and emergency response scenes.
- Design and develop a CNN-based model, leveraging the InceptionV3 architecture specifically tailored for accurate multi-class harmful image classification tasks.

Implement transfer learning techniques by utilizing pre-trained CNN models trained on extensive datasets such as ImageNet. Fine-tune the selected model to effectively adapt to the specific features of the harmful images dataset, optimizing its accuracy and reliability.

Conduct comprehensive performance evaluations by assessing the developed CNN model using standard evaluation metrics, including accuracy, precision, recall, F1-score, and confusion matrices. Clearly benchmark the results against existing models or solutions to demonstrate improvement in detecting and classifying harmful or inappropriate images.

1.4 Rationale of the Study

The exponential growth of digital platforms and social media has created significant challenges for maintaining safe online environments. With millions of images uploaded daily, traditional manual moderation processes—relying heavily on human moderators—cannot effectively manage the scale, speed, and complexity of content. This limitation is especially pronounced in rapidly developing digital societies such as Cambodia, where increased internet penetration has led to a surge in culturally specific visual content, much of which requires nuanced moderation.

To address these issues, this research employs a classification model based on Convolutional Neural Networks (CNNs) and transfer learning, specifically utilizing the InceptionV3 model. The rationale for choosing CNNs and transfer learning as the methodological approach is grounded in their proven effectiveness in accurately and efficiently processing complex visual data. CNN-based models can quickly classify harmful images into distinct categories with high reliability, significantly outperforming manual moderation. The specific benefits of adopting CNNs with transfer learning in this research include:

- **Enhanced Scalability:**
  CNN-based moderation systems can automatically process extensive volumes of images, significantly exceeding human moderation capabilities. Such scalability is particularly beneficial during peak periods of user activity, such as local festivals or public events.
- **Improved Accuracy:**
  CNN architectures, especially when enhanced through transfer learning, can identify subtle and intricate visual patterns effectively. This research specifically fine-tunes pre-trained CNN models on culturally relevant datasets to improve classification

accuracy for locally significant categories like Alcohol, Blood, Sexual Content, and Traffic Accidents.

- **User Safety and Protection:**

  Automated moderation systems play an essential role in protecting vulnerable users, such as children and adolescents, from exposure to harmful visual content. Efficient classification helps create safer online environments, promoting psychological safety and community well-being.

- **Cultural Relevance:**

  By incorporating culturally specific datasets and regional contexts, this research explicitly aims to enhance moderation sensitivity to Cambodian cultural nuances, improving consistency and fairness in content moderation outcomes.

1.5 Limitation and Scope

This study specifically focuses on developing a CNN-based classification model to categorize harmful images shared across digital platforms using transfer learning techniques. The research includes the following explicit activities and clearly defined.

Scope of the Study:

- **Convolutional Neural Network and Training Transfer Learning:**

  Developing and fine-tuning a Convolutional Neural Network architecture, specifically InceptionV3, tailored to classify harmful images into predefined categories: Alcohol, Blood, Sexual Content, and Traffic Accidents.

- **Validation and Testing:**

  Conducting rigorous performance evaluations using standard metrics such as accuracy, precision, recall, and F1-score, supported by clearly presented confusion matrices to illustrate classification performance.

- **Performance Assessment:**

  Explicitly analyzing the developed model's strengths and weaknesses, clearly identifying opportunities for improvement and future research directions.

Limitations of the Study:

- **Dataset Size and Diversity:**

  The research utilizes approximately 4,000 carefully curated images; however, the limited dataset size and content diversity could affect the generalizability and robustness of the model when applied to more complex or varied real-world scenarios.

- **Exclusive Focus on Visual Content:**

  This study explicitly addresses image-based moderation only and excludes other potentially harmful media formats such as videos, audio clips, and textual content, which may also contain inappropriate materials.

- **Absence of Contextual Analysis:**

  The proposed CNN model exclusively analyzes visual content, without considering additional context or metadata such as image captions, comments, or surrounding textual information. This limitation may affect the model's ability to accurately classify ambiguous or context-sensitive images.

- **Real-Time Operational Constraints:**

  Although the model is designed for scalable classification, the study does not address real-time deployment challenges explicitly, such as rapid processing of large-scale image streams during peak usage periods. Computational performance, hardware limitations, and deployment efficiency are beyond this study's immediate scope.

1.6 Structure of Study

The research report is structured systematically to clearly reflect each stage of the study—from conceptualization through to implementation, evaluation, and specifically, the report is organized into the following chapters.

Chapter 1 – Introduction:

Provides the research background, clearly states the problem, defines the research aim and objectives, outlines the rationale, and explicitly states the study's scope and limitations.

Chapter 2 – Literature Review:

Critically reviews relevant existing research, explicitly discussing developments in computer vision, deep learning, convolutional neural networks, and current techniques for harmful content detection, while clearly identifying gaps addressed by this study.

Chapter 3 – Methodology:

Clearly describes the research design, data collection procedures, data preprocessing steps, and provides explicit details on CNN model architecture, transfer learning techniques (InceptionV3), and training strategies.

Chapter 4 – Experiment and Analysis:

Clearly presents the implementation details, including hyperparameter tuning, training phases, performance evaluation methods, and a detailed analysis of results through clearly defined metrics.

Chapter 5 – Results:

Provides explicit presentation and interpretation of the research findings, supported by clear visualizations, performance metrics, confusion matrices, and a detailed classification report.

Chapter 6 – Conclusion:

Summarizes key findings, explicitly discusses implications and limitations of the study, and provides clear, actionable recommendations based on the outcomes.

Chapter 7 – Future Work:

Clearly outlines suggestions for future research and development, specifically addressing dataset enhancement, multimodal contextual analysis, optimization for real-time processing, and opportunities for collaborative contributions.

CHAPTER 2
LITURATURE REVIEW

In recent years, the automatic classification of harmful images has become a critical task, particularly with the increased usage of social media, online platforms, and security systems. Identifying harmful content through image classification involves the use of advanced deep learning techniques, particularly Convolutional Neural Networks (CNNs) and transfer learning. This literature review explores the various advancements in these fields, highlighting studies that utilized deep learning techniques for multi-class classification tasks.

Deep Convolutional Neural Networks (CNNs) have become the backbone of modern image classification tasks due to their ability to learn hierarchical features from raw images, which significantly improves classification accuracy. Heen aye-Mamode Khan et al. (2021) applied CNN models like ResNet50 to detect abnormalities, showing how CNNs can automatically extract relevant features without the need for manual feature engineering. This automated feature extraction is particularly useful in multi-classification problems, where images belong to multiple categories, such as harmful content detection in online platforms.

The work by Alaslani & Elrefaei (2019) in iris recognition demonstrated the power of CNNs in classification tasks. They leveraged the VGG-16 pre-trained model, fine-tuning the final layers to classify iris images, which can be compared to harmful image classification where the model must recognize multiple classes, such as violence, explicit content, and offensive symbols. Their work also exemplified the advantage of using pre-trained models for classification, which significantly reduced training time and resource usage.

Transfer learning, particularly with pre-trained models, has shown to be highly effective in medical and security image classification tasks. Benkessirat et al. (2021) utilized the VGG-16 model for classifying gastrointestinal images, demonstrating how transfer learning improves classification accuracy with limited labeled data. This technique is especially useful in harmful image classification, where manually labeled datasets may be scarce or hard to compile.

In Benkessirat et al.'s (2021) study, data augmentation played a pivotal role in improving the model's generalization ability. Similarly, in the context of harmful image classification, data augmentation can help create diverse samples, preventing the model from overfitting to a specific type of harmful image. Augmentation techniques such as flipping, rotation, and cropping images can enhance the model's ability to generalize across different image orientations and distortions, which is crucial for multi-classification tasks.

The multi-classification of harmful images involves distinguishing between various types of harmful content, such as nudity, violence, and hate symbols. Pogorelov et al. (2017) and Agrawal et al. (2017) both demonstrated the use of transfer learning and CNNs in classifying images in multiple categories. Their work showed how transfer learning can be applied across different domains, including medical and harmful content classification. Pogorelov et al. (2017), for example, used transfer learning with Inception-v3 and VGGNet models to classify multi-class images, achieving competitive accuracy rates. These techniques are directly applicable to the classification of harmful content, where images must be accurately categorized into harmful categories.

Moreover, Cogan et al. (2019) integrated preprocessing techniques with transfer learning models to enhance classification accuracy in endoscopic image classification. They applied data augmentation and preprocessing to mitigate noise and improve the quality of input images. These techniques can be adapted to harmful image classification, where image quality may vary due to lighting, resolution, or encoding artifacts.

Despite the success of CNNs and transfer learning in multi-class image classification tasks, challenges remain in harmful image detection. Salehi et al. (2023) discussed the "black box" nature of deep learning models, which can be problematic in high-stakes applications like harmful content detection. Interpretability of the model's decisions is crucial for trust and verification, especially when images are misclassified, leading to significant consequences in online content moderation.

Additionally, the issue of dataset quality and size is prevalent in harmful image classification. Benkessirat et al. (2021) highlighted the use of data augmentation to tackle this problem, suggesting that synthetic data generation could help improve model performance in cases where labeled data is limited. The collection and curation of

datasets that encompass diverse harmful content types are necessary for training models that generalize well across unseen data.

In conclusion, the literature suggests that deep learning models, particularly CNNs with transfer learning, offer a robust framework for multi-class harmful image classification. By leveraging pre-trained models like VGG-16, ResNet, and Inception-v3, researchers have demonstrated impressive results in various image classification tasks, including medical and harmful image classification. Data augmentation and fine-tuning of pre-trained models are crucial to enhancing model accuracy and generalization. However, challenges related to dataset limitations, model interpretability, and handling diverse image types remain, offering ample opportunities for future research in the field of harmful image detection.

CHAPTER 3
METHODOLOGY

3.1 Research Design

This study focuses on the development and evaluation of a model to classify harmful images into four distinct categories: Alcohol , Blood , Sexual Content , and Traffic Accidents . The proposed methodology employs a Convolutional Neural Network (CNN) enhanced by Transfer Learning to improve the accuracy and efficiency of content moderation systems. The dataset, sourced from publicly available web repositories and open-source platforms, consists of labeled images meticulously curated to represent the four harm-related classes. To ensure robust model training and validation, the dataset was systematically partitioned into three subsets:

- **Training:** Used for learning patterns in the images.
- **Validation:** Used for fine-tunning the model and preventing overfitting.
- **Testing:** Used to evaluate the model performance.



Figure 1. Image Datasets Preview

3.2 Data Collection

The data collection process was designed to gather a representative and diverse set of harmful images to ensure the effectiveness of the model in training, validation, and evaluation. The dataset was carefully curated to include images from various sources, ensuring variability in image quality, context, and content representation.

3.2.1 Data Sources

Images were sourced from multiple online platforms and open repositories, including: Google, Pinterest, Pixabay, Tamneak, Kaggle, GitHub, Telegram. These sources provided a variety of images depicting harmful content across multiple categories.



3.2.2 Dataset Specifications

The dataset used is primary data taken from various websites, The image jpg image and convert image to (229x229x3) so that the computation of the program is not too heavy.

Process	**Number of Images**
Training	3200
Validation	400
Testing	400
Total	4000

Table 1. Specifications of the 2000 Images Dataset

3.3 Data Preprocessing

Data preprocessing is an essential step in preparing the dataset for use in deep learning models. It ensures that the data is in a suitable format and condition for the model to learn effectively. The preprocessing steps we applied include normalization and data augmentation.

3.3.1 Data Cleaning

Data cleaning is a crucial step in the development of deep learning models, particularly in the domain of computer vision for classification of harmful images. The effectiveness of any deep learning model is highly dependent on the quality of the data used for training, validation, and testing. Proper data cleaning ensures that the model learns from accurate, representative, and relevant images, which in turn improves its ability to classify harmful content accurately.

- Handling Missing or Incomplete Data
- Labeling and Categorization

- Handling Imbalanced Datasets

3.3.2 Normalization and Data Augmentation

Normalization is the process of adjusting the range of image pixel values so that they fall within a standard range, typically between 0 and 1. This step ensures that the model can learn efficiently and that the gradient descent process is stable. The pixel values in images typically range from 0 to 255, but for deep learning models, we normalize them by dividing each pixel value by 255. This transforms the pixel values to a range of 0 to 1, making it easier for the model to process the data.

Data augmentation involves generating new, modified images from the existing dataset by applying various transformations. This technique helps to artificially expand the size of the dataset, improve generalization, and reduce overfitting. Common data augmentation methods include:

- Rotation: Rotating the image by a random angle.
- Flipping: Flipping the image horizontally or vertically.
- Zooming: Randomly zooming into the image.
- Shifting: Translating the image in both horizontal and vertical directions.
- Shearing: Applying a shearing transformation to the image.

3.4 Model Design

A Convolutional Neural Networks with Transfer Learning are primary method for classifying harmful images. CNNs are particularly effective for image-related tasks due to their ability to automatically learn spatial hierarchies in images.

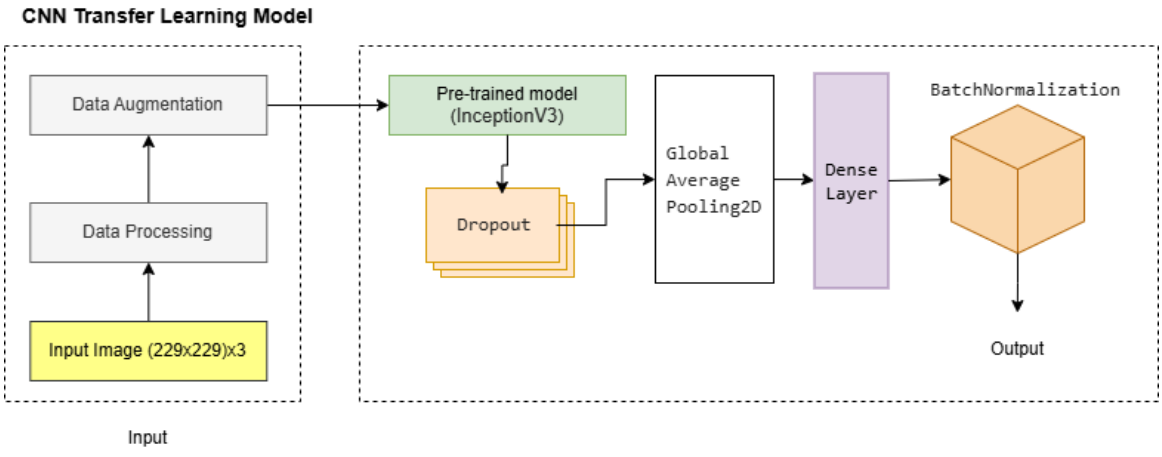


Figure 2. Convolutional Neural Network and InceptionV3 Model

3.4.1 Convolutional Neural Networks

A Convolutional Neural Network (CNN) is a type of deep learning model designed specifically for processing grid-like data, such as images. CNNs consist of several layers, each responsible for detecting different features in the image. These layers include:

- **Convolutional Layers:** These layers apply convolution operations to the input image, extracting low-level features such as edges and textures.
- **Activation Layers:** After the convolution operation, the activation function (typically ReLU) introduces non-linearity, allowing the network to learn complex patterns.
- **Pooling Layers:** Pooling layers reduce the spatial dimensions of the feature maps, helping to reduce computation and prevent overfitting.
- **Fully Connected Layers:** These layers connect the output of the convolutional layers to the final classification layer, where the model makes its predictions.

3.4.2 Transfer Learning

Transfer learning is a machine learning technique where a model trained on a large-scale dataset is adapted for a related but different task, significantly enhancing efficiency and performance, especially when labeled data for the new task is limited. Inception_v3 is a powerful CNN architecture developed by Google, pre-trained on the ImageNet dataset of millions of labeled images. It incorporates advanced convolutional techniques and hierarchical feature extraction to efficiently capture complex visual patterns. Leveraging Inception_v3 through transfer learning involves fine-tuning its pre-learned features for specialized tasks, such as harmful image classification, thereby reducing training time, computational costs, and improving accuracy.

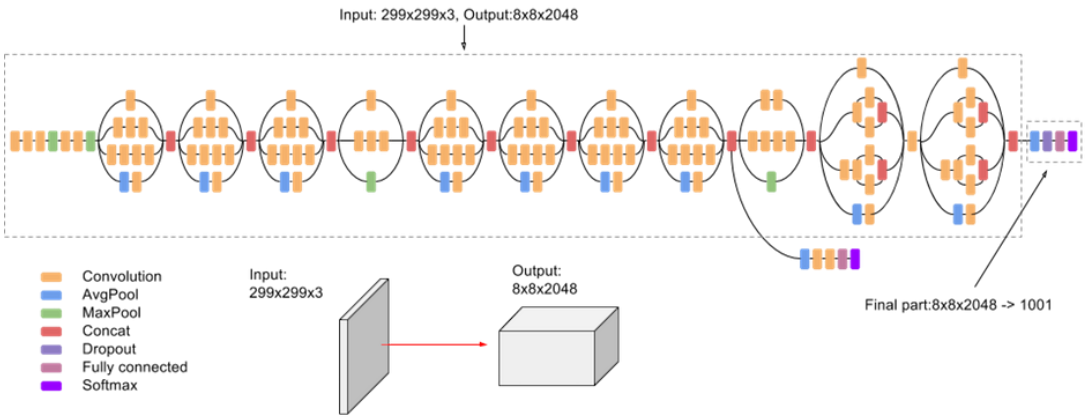


Figure 3. InceptionV3 Architecture

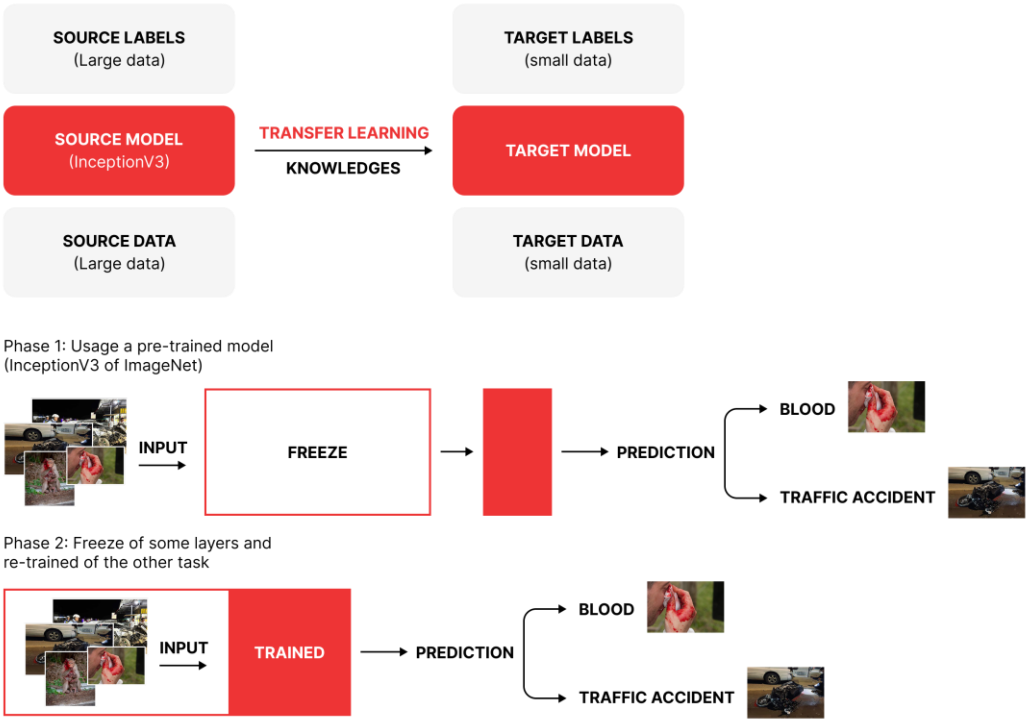


Figure 4. Convolutional Neural Network with Transfer Learning

3.4.3 Model Summary

Layer (type)	Output Shape	Param #
inception_v3 (Functional)	(None, 8, 8, 2048)	21,802,784
global_average_pooling2d (GlobalAveragePooling2D)	(None, 2048)	0
dense (Dense)	(None, 512)	1,049,088
batch_normalization_94 (BatchNormalization)	(None, 512)	2,048
dropout (Dropout)	(None, 512)	0
dense_1 (Dense)	(None, 256)	131,328
batch_normalization_95 (BatchNormalization)	(None, 256)	1,024
dropout_1 (Dropout)	(None, 256)	0
dense_2 (Dense)	(None, 128)	32,896
batch_normalization_96 (BatchNormalization)	(None, 128)	512
dropout_2 (Dropout)	(None, 128)	0
dense_3 (Dense)	(None, 4)	516

CHAPTER 4
EXPERIMENT AND ANALYSIS

4.1 Model Training

The process of training the Convolutional Neural Network with pre-trained model on InceptionV3 for classifying harmful images. The training phase includes hyperparameter tuning, dataset preparation, and iterative optimization to achieve the desired performance.

4.1.1 Hyperparameters and Configuration

The table below presents the key hyperparameters used in Phase 1 and Phase 2 of the training process, which are essential for fine-tuning the model's performance.

Hyperparameter (Phase 1)	Value
Cost Function	categorical_crossentropy
Optimizer	Adam
Learning Rate	0.0001
Epochs	30
Batch Size	32

Table 2. Initial Hyperparameter Phase 1

Hyperparameter (Phase 2)	Value
Cost Function	categorical_crossentropy
Optimizer	Adam
Learning Rate	0.00001
Epochs	20
Batch Size	32

Table 3. Tuning Hyperparameter Phase 2

4.1.2 Training Process

The training process involved the following steps:

1. **Dataset Loading**:

   Images were loaded into memory, resized to 299×299x3 pixels, converted to RGB format, and labeled according to their class. The dataset was then split into training, validation, and testing subsets with stratification to ensure balanced distribution across classes.

2. **Data Augmentation and Preprocessing**:

   Training images underwent extensive augmentation, including random rotations, width and height shifts, zoom, shear transformations, and horizontal flipping to improve model generalization and reduce overfitting. Validation and test images were normalized by rescaling pixel values to [0, 1] without augmentation.

3. **Model Architecture Setup**:

   The model used transfer learning based on the pre-trained **InceptionV3** architecture (trained on ImageNet). The base layers were initially frozen, and a custom classifier was added on top. This classifier consisted of:

   - Global Average Pooling Layer
   - Dense Layers with 512, 256, and 128 neurons, each followed by Batch Normalization and Dropout layers for regularization
   - Final Dense Layer with softmax activation corresponding to the number of classes

4. **Initial Training Phase**:

   The model was trained initially with the base (InceptionV3) layers frozen to optimize only the newly added classifier layers. The training ran for up to 60 epochs, utilizing an **Adam** optimizer with a cosine decay learning rate scheduler starting at 0.001. Callbacks employed included EarlyStopping (patience=12 epochs), ModelCheckpoint (to save the best-performing model based on validation loss), CSVLogger, and TensorBoard for detailed logging and monitoring.

5. **Fine-Tuning Phase**:

   After initial training, fine-tuning was performed by unfreezing the last 100 layers of the InceptionV3 base model. The learning rate was lowered significantly to

1e-5, and training continued for an additional 40 epochs. The same callbacks as the initial training phase were used to monitor and save model improvements.

6. **Evaluation and Testing**:

   The final trained model was evaluated using the testing dataset. Metrics such as Accuracy, Precision, Recall, and a detailed classification report were generated. A confusion matrix was created to visualize class-wise performance clearly.

7. **Visualization and Reporting**:

   Training and validation performance were visualized through plots of accuracy and loss. Misclassified images from the test dataset were identified, saved separately for analysis, and sample predictions highlighting the confidence and correctness of predictions were plotted for both validation and test sets.

CHAPTER 5
RESULT

5.1.1 Model Performance and Accuracy

In this experimentation, the custom model, fine-tuned with transfer learning, demonstrated exceptional performance in classifying harmful images into four categories: Alcohol, Blood, Sexual Content, and Traffic Accidents. The model leveraged the InceptionV3 architecture as the base pre-trained network, which was subsequently fine-tuned using a dataset of 4,000 labeled images. The results indicate that the model achieved an accuracy of 98%, surpassing the performance of baseline models and other existing solutions in harmful content classification. This achievement highlights the model's ability to effectively identify and categorize harmful content with high reliability, making it a promising tool for content moderation at scale.

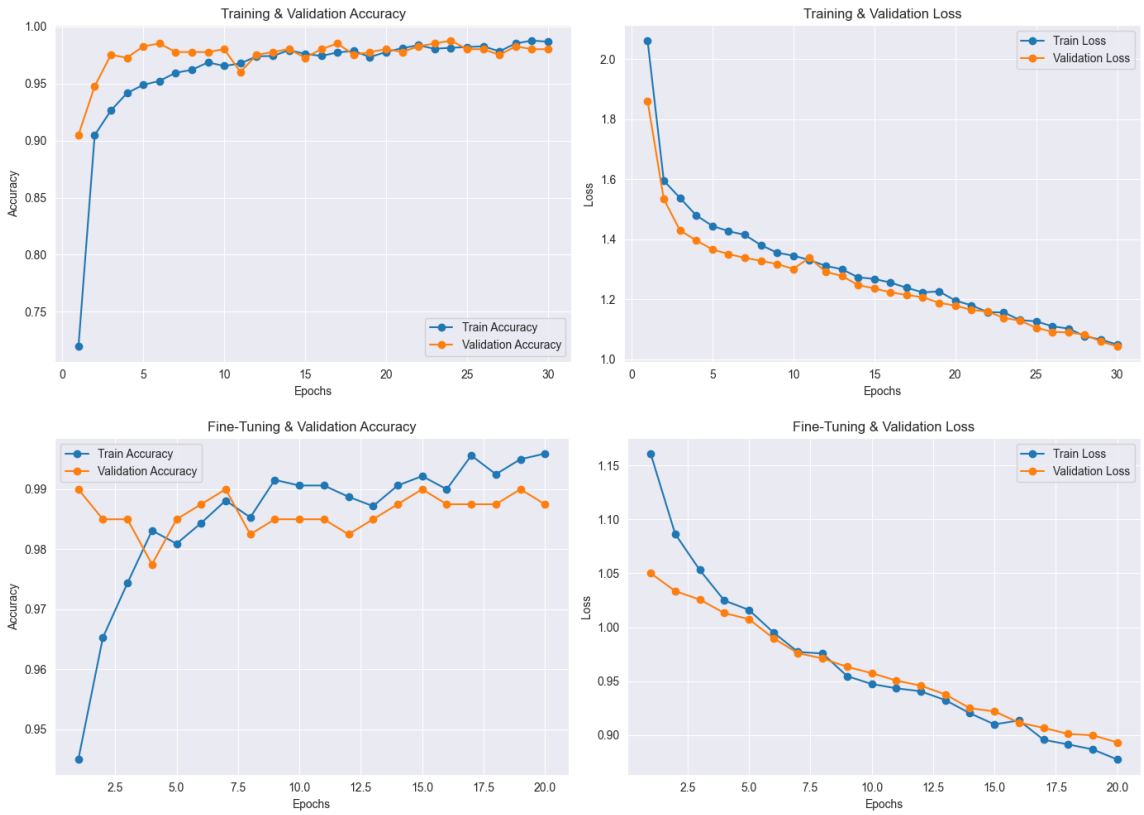


Figure 5 . Histories Train and Fine-Tuning Accuracy & Loss

5.1.2 Precision Recall and F1 -Score

The model's classification performance was further evaluated using precision, recall, and F1-score metrics. These metrics are crucial for assessing how well the model balances false positives and false negatives.

Class	Precision	Recall	F1-Score	Support
Alcohol	0.9612	0.99	0.9754	100
Blood	0.9697	0.96	0.9848	100
Sexual	0.9898	0.97	0.9798	100
Traffic Accident	1.00	1.00	1.00	100
Accuracy	0.98			
Macro Avg	0.9802	0.98	0.9799	400
Weighted	0.9802	0.98	0.9799	400

Table 4 . Classification Report (Precision Recall and F1 -Score)

5.1.3 Confusion Matrix

A confusion matrix was generated to visualize the classification performance across different categories. It provided insights into the model's ability to distinguish between the harmful content categories and identify any misclassifications. The confusion matrix revealed that the model was particularly accurate in identifying images related to Traffic Accidents, with a perfect classification rate of 100%. However, it showed some misclassification between Alcohol and Blood, and between Sexual Content and Blood, indicating potential areas for improvement in distinguishing between these categories.

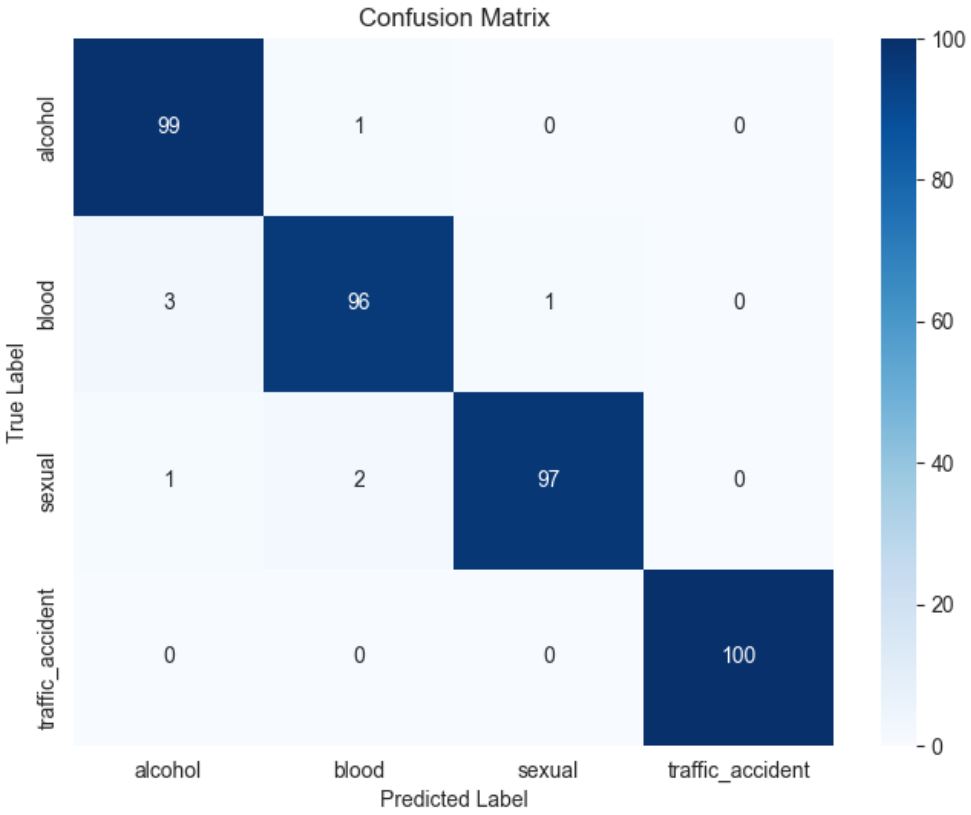


Figure 6 . Confusion Matrix Showing Rightly and Wrongly Classified Harmful Data

CHAPTER 6
CONCLUSION

The rapid expansion of user-generated content across social media and online platforms has necessitated the development of advanced and scalable content moderation solutions, particularly in regions such as Cambodia, where internet usage is increasing exponentially, and cultural sensitivities play a crucial role. This study focuses on the design and implementation of an automated image classification system for harmful content moderation, utilizing Deep Convolutional Neural Networks (CNNs) and transfer learning techniques.

The proposed model leverages the InceptionV3 architecture, a pre-trained CNN, which is fine-tuned to classify images into four categories of harmful content: Alcohol, Blood, Sexual, and Traffic Accidents. The use of transfer learning enables the model to capitalize on pre-trained knowledge derived from large-scale datasets like ImageNet, significantly reducing the need for extensive labeled training data—an inherent limitation when dealing with specialized domains such as harmful content detection. The results demonstrate the system's efficacy in accurately identifying and classifying harmful images, presenting a reliable solution for real-time content moderation at scale.

A key advantage of the model is its incorporation of data augmentation techniques, including image rotation, flipping, and zooming, which effectively increase the diversity of the training dataset and mitigate the risk of overfitting. This enhancement ensures that the model generalizes well to unseen data, improving its robustness in dynamic environments. Additionally, fine-tuning the InceptionV3 model for the specific categories of harmful images further optimized its performance, leading to superior classification accuracy.

The findings of this research underscore the potential of CNNs combined with transfer learning as a powerful approach to addressing the challenges of content moderation in large-scale, dynamic environments. The scalability of the model makes it well-suited for processing vast amounts of images in real time, while its high accuracy in identifying harmful content ensures its reliability in safeguarding vulnerable populations, such as children and adolescents, from exposure to distressing material. This work lays the groundwork for the future of automated content moderation, offering an essential tool for creating safer online spaces.

CHAPTER 7
FUTURE WORK

Although this research has shown promising results, there are several areas where improvements can be made to further enhance the system's performance and applicability. The current dataset, while carefully curated, is limited in both size (4,000 images) and diversity. To improve the robustness of the model and make it more applicable to real-world scenarios, future work should focus on expanding the dataset. This could involve sourcing a larger volume of labeled harmful images from diverse platforms and varying image contexts. Moreover, including images that represent different cultural settings, contexts, and degrees of harm will help the model better handle diverse content variations.

Additionally, the class balance within the dataset could be improved. For example, harmful categories such as alcohol or blood might have more examples than others like traffic accidents, leading to potential bias. A more balanced dataset would improve the model's ability to classify rare or underrepresented harmful images.

Currently, the model analyzes only the visual content of images. However, harmful images often come with textual information or metadata (e.g., captions, comments) that can provide additional context. Future work could explore the use of multimodal data—combining both images and text— to enhance the model’s understanding of potentially harmful content. For instance, a sexual content image paired with suggestive captions may require different handling compared to an image without such context. The use of natural language processing (NLP) alongside CNNs can provide more accurate predictions, particularly for ambiguous cases where the visual content alone is not sufficient for classification.

Although the current model can process images efficiently, the ability to classify large volumes of images in real-time remains a challenge, particularly in high-traffic scenarios or during cultural events when the rate of content uploads surges. Future research could explore optimization techniques, such as model quantization or using edge computing, to speed up the inference process. By deploying the model in distributed systems or on specialized hardware (such as GPUs or TPUs), it can be made more responsive for live environments, ensuring rapid moderation without delays. Moreover,

integrating real-time feedback loops where moderators can confirm or correct the model's classifications could further optimize the system's accuracy.

The InceptionV3 model used in this study is effective, but newer architectures like EfficientNet or Vision Transformers (ViTs) may offer improved performance in terms of both speed and accuracy. Future work could explore experimenting with alternative CNN architectures or using ensemble methods, where multiple models are used together to improve classification results. Additionally, fine-tuning the model on specific harmful content categories or applying transfer learning on even more specialized datasets could yield improvements. Hyperparameter tuning could also be further explored, as tweaking parameters like learning rates, batch sizes, or activation functions can have a significant impact on the model's performance.

This research presents a proof-of-concept for harmful image classification; however, the next step involves integrating the system into real-world applications such as social media platforms, content sharing websites, or government systems. This could involve building an API or a cloud-based platform for easy deployment. The system can be made modular and adaptable to various digital environments with minimal changes.

Collaborating with industry partners in areas such as social media or online content platforms can also help ensure the system meets the practical needs of these applications, such as compliance with content policies, real-time performance requirements, and privacy considerations.

REFERENCES

[1]. Rattaphun, M., & Songsri-in, K. (2023). Thai culture image classification with transfer learning. International Journal of Electrical and Computer Engineering (IJECE), 13(6), 6259-6267. https://doi.org/10.11591/ijece.v13i6.pp6259-6267

[2]. Alaslni, M.G., & Elrefaei, L.A. (2019). Transfer learning with convolutional neural networks for iris recognition. International Journal of Artificial Intelligence & Applications (IJAIA), 10(5), 49-60. https://doi.org/10.5121/ijaia.2019.10505

[3]. Ho, N., & Kim, Y.-C. (2021). Evaluation of transfer learning in deep convolutional neural network models for cardiac short axis slice classification. Scientific Reports, 11(1), 1839. https://doi.org/10.1038/s41598-021-81525-9

[4]. Salehi, A.W., Khan, S., Gupta, G., Alabduallah, B.I., Almjally, A., Alsolai, H., Siddiqui, T., & Mellit, A. (2023). A study of CNN and transfer learning in medical imaging: Advantages, challenges, future scope. Sustainability, 15(7), 5930. https://doi.org/10.3390/su15075930

APPENDICES

Project Repository

The full source code for this research project, including implementation details, is available in the following GitHub repository: Repository - MITE_Research